

Size-sensitive Sequence of Tense in Hungarian ¹

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Abstract

The English sentence *John said that Mary was pregnant* is ambiguous: Its *simultaneous* reading is *John said: ‘Mary is pregnant’*; its *backshifted* reading is *John said: ‘Mary was pregnant’*. Under Sequence of Tense Analyses of such *past-under-past* configurations (e.g. Ogihara 1995), the simultaneous reading arises if an embedded past head undergoes “deletion”-under-Agree with the matrix past head (the embedded past form is phonetically unaffected). Without deletion-under-Agree, the embedded past tense remains backshifted. The availability of deletion-under-Agree has been handled as a language-wide parameter, e.g. unlike English, Japanese lacks deletion-under-Agree (e.g. Ogihara and Sharvit 2012). In Hungarian, *speech-reporting* clauses are Japanese-like and obligatorily backshifted, whereas *non-speech-reporting* clauses are English-like, i.e. have a simultaneous reading. I propose that the impossibility of deletion-under-Agree in speech-reporting clauses comes from *size-dependent opacity* (e.g. Williams 2003).

Keywords: Williams Cycle, size-dependent opacity, Sequence of Tense, Hungarian

1 Introduction

In English, the matrix and the embedded past tenses can receive a *simultaneous* reading:

In (1), John can report Mary’s pregnancy at the time (i.e. *John said: “Mary is pregnant.”*).

(1) *John said Mary was pregnant.*

According to Sequence of Tense analyses (Ogihara 1995; Abusch 1997), the simultaneous reading arises because the embedded *PAST* is “deleted” once it agrees with the

c-commanding matrix *PAST* head as in (2). The deleted *PAST* in (2) is interpreted as a non-indexical present relative to the matrix past tense.

- (2) [*PAST* t_0] λ_1 [*John* [*say* t_1] λ_2 [[~~*PAST*~~ t_2] λ_3 [*Mary* [*be* t_3] *pregnant*]]]

If no Sequence of Tense rule applies, the embedded clause remains *backshifted*; in the context of (1), John makes a report of a pregnancy preceding his “now” (i.e. *John said: “Mary was pregnant.”*). According to Sequence of Tense approaches (for a survey see Sharvit 2020), languages like Japanese lack this Sequence of the Tense Rule entirely, i.e. their past-under-past configurations have only backshifted readings (Ogihara and Sharvit 2012).

The empirical claim of this paper is that Hungarian embedded clauses can be divided into two classes: Embedded clauses not encoding the content of some speech act work like English embedded clauses in that their past can receive a simultaneous reading with the matrix past. For example, the sentence in (3) can mean that Mary was sad *at the time* in my dream.²

- (3) *Az-t álmod-t-am,* [*hogy Mari szomorú vol-t-Ø*].
 it-ACC dream-PST-1SG.DEF that Mari sad be-PST-3SG.INDF
 ‘I dreamed the following: “Mary is sad”.’ (backshifted reading set aside)

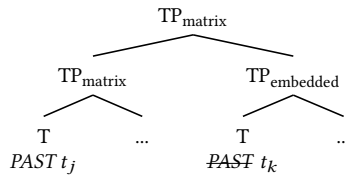
In contrast, Hungarian embedded clauses encoding the content of some speech act work like embedded clauses in Japanese. Hence, the embedded time in (4) is obligatorily backshifted relative to the matrix time (i.e. the sayer must report Mary’s sadness before the shouting time).

- (4) *Az-t mond-t-am,* [*hogy Mari szomorú vol-t-Ø*].
 it-ACC say-PST-1SG.DEF that Mari sad be-PST-3SG.INDF
 ‘I said: “Mary was sad”.’ but # ‘I said: “Mary is sad”.’

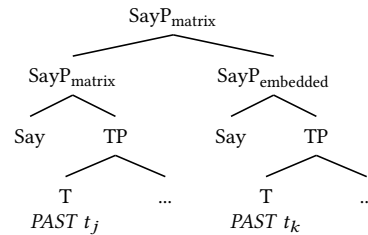
In my analysis, whether a clause denotes a speech-report or not is primarily determined by the internal structure of the clause, and what matrix environments the various clauses

can occur in is decided by the pragmatic context. Assuming that *hogy* ‘that’ does not have a dedicated CP projection in Hungarian (Egressy 2023, 2025), speech-reporting and non-speech-reporting clauses are of different sizes: Non-speech-reporting clauses are TPs in (5), while speech-reporting clauses are SayPs as in (6), where Say>T in the functional sequence. Furthermore, I follow the spirit of Williams’ (2003) Level Embedding Conjecture by assuming that Hungarian embedded clauses attach to matrix projections of their own sizes: TP to TP as in (5), SayP to SayP (6). The final component of the analysis says that the Sequence of Tense deletion rule applies in the cyclic structure building of *narrow syntax*, in the spirit of Cyclic Agree (Rezac 2003; Béjar and Rezac 2009): It can apply iff a projection of the matrix *PAST* contains the embedded *PAST* (i.e. c-command is not necessary). Therefore, the “deletion” of the embedded *PAST* is possible in the TP-TP adjunction in (5) because the highest matrix TP projection of the matrix *PAST* contains the embedded *PAST* but not in the SayP-SayP adjunction in (6) because no projection of the matrix *PAST* contains the embedded *PAST*. Consequently, (5) has a simultaneous reading while (6) does not. In general, the phenomenon interpretable an instance of the Williams Cycle (Williams 1974, 2003, 2013; Poole 2023; Meadows 2023; Egressy 2023, 2025) or Agree-based *size-dependent* opacity (Keine 2016, 2019, 2020) in the realm of Sequence of Tense: The deletion rule initiated from T cannot see through projections headed by the Say head because SayPs are structurally too large (formally: Say>T in *fseq*).

(5)



(6)



The discussion proceeds as follows: In Section 2, I present the empirical data on the two types of embedded clauses. Section 3 describes a particular implementation of the

Sequence of Tense rule as it operates in other languages. In Section 4, I propose the syntactic analysis which combines size-dependent attachment with the Cyclic Sequence of Tense (sketched above). Due to space limitations, I do not include a semantic analysis in the main text, but interested readers are welcome to take a look at the Appendix for the sketch of a semantic analysis.

2 The two classes of Hungarian clauses

2.1 Non-speech-reporting clauses have a simultaneous reading

This subsection discusses various Hungarian embedded clauses introduced by *hogy* ‘that’ *not* describing the content of speech acts. Clauses with the same internal structure can occur in a number of environments: They can be arguments of verbs of direct perception, dreaming, thought, belief; furthermore, these such clauses can also occur in non-argument positions. The main generalization of the section is that independently of their environment (argumenthood/adjuncthood), clauses in this class have a *simultaneous* reading in past-under-past configurations. In other words, this set of clauses exhibits the general behaviour of English *that*-clauses (as discussed in the Introduction and Section 3).

Hungarian argument clauses introduced by *hogy* ‘that’ are always embedded with the expletive *az* ‘it’, which is case-marked according to grammatical function of the embedded clause (Kenesei 1994; É. Kiss 2002). Hence, the expletive of the object clause of *lát* ‘see’ in (7) is marked by accusative. Nominative- and accusative-marked expletive can be pro-dropped in its post-verbal position, however, its presence is indicated by the definite agreement on the verb (for a more extensive discussion, see Egressy 2023).

The sentence (7) exemplifies the *past-under-past* configuration: The embedded past verb *volt* ‘was’ is embedded under a matrix past verb *látta* ‘saw’. Importantly, the sentence in (7) describes a scene of direct perception: Peti sees the event encoded in the embedded

clause as it is happening *at the same time*. Since this past-under-past configuration is compatible with the direct-perception scene, we can declare that the past-under-past configuration in (7) has a *simultaneous* reading.

- (7) *Peti a saját szem-é-vel lát-t-a (az-t), [hogy Mari szomorú*
 Peti the own eye-3SG.POSS-INS see-PST-3SG.DEF it-ACC that Mari sad
vol-t-Ø].
 be-PST-3SG.INDF
 ‘Peti saw the following with his own eyes: “Mary is sad”.’
 (apparent backshifted reading discussed in Section 2.2)

Speakers of Hungarian may claim that some of the *past-under-past* sentences of the section also have a *backshifted* reading. However, the *backshifted* readings for these sentences have a flavour of indirectness. For instance, the backshifted reading of (7) must denote seeing some sign of Mary’s sadness in the past, e.g. some paper towels that she used while crying. While this raises a number of questions, the backshifted reading of non-speech-reporting clauses in past-under-past configurations will be discussed alongside the past-under-past configurations of speech-reporting clauses in the next subsection.

The behaviour of *lát* ‘see’ applies to all verbs of direct perception, hence, the complement clause of *hall* ‘hear’ also has a simultaneous reading in past-under-past configurations (for a more feasible scenario, I use a slightly different embedded sentence):³

- (8) *Peti a saját fül-é-vel hall-ott-a (az-t), [hogy Mari*
 Peti the own ear-3SG.POSS-INS hear-PST-3SG.DEF it-ACC that Mari
sír-t-Ø].
 cry-PST-3SG.INDF
 ‘Peti heard the following with his own ears: “Mary is crying”.’ (Peti heard the noise of crying; The apparent backshifted reading discussed in Section 2.2)

Dreaming can also be handled as an example of direct perception (“seeing in a dream”). Correspondingly, the sentence in (9) with the matrix verb *álmodik* ‘dream’ has a simultaneous reading in a past-under-past configuration.

- (9) *Peti az-t álmod-t-a, [hogy Mari szomorú vol-t-Ø].*
 Peti it-ACC dream-PST-3SG.DEF that Mari sad be-PST-3SG.INDF
 ‘Peti saw the following in his dream: “Mary is sad”.’
 (apparent backshifted reading discussed in Section 2.2)

Complement clauses to verbs of thought and belief also belong to the class of non-speech-reporting clauses: It is possible to have thoughts about the events of the world without putting thoughts into a verbal message. Thus, the past-under-past sentence in (10) with matrix verbs *hisz* ‘believe’ or *gondol* ‘think’ has a simultaneous reading.

- (10) *Peti az-t {gondol-t-a/ hit-t-e}, [hogy Mari szomorú*
 Peti it-ACC think-PST-3SG.DEF believe-PST-1SG.DEF that Mari sad
vol-t-Ø].
 be-PST-3SG.INDF
 ‘Peti {thought/believed} the following: “Mary is sad”.’
 (apparent backshifted reading discussed in Section 2.2)

Non-speech reporting clauses can correspond to the subject of the matrix clause, too. In example (11), since the verb *aggaszt* takes a subject clause, it’s embedded with the nominative expletive *az*. However, similar to the object clauses listed above, the past-under-past configuration has simultaneous reading.

- (11) *Az aggaszt-ott-a Peti-t, [hogy Mari szomorú vol-t-Ø].*
 it worry-PST-3SG.DEF Peti-ACC that Mari sad be-PST-3SG.INDF
 ‘The following worried Peti: “Mary is sad”.’
 (apparent backshifted reading discussed in Section 2.2)

Finally, non-speech-reporting clauses do not have to be arguments. As É. Kiss (2002) notices, *hogy*-clauses having the same form as argument clauses can co-occur with intransitive matrix verbs.⁴ Consider example (12) with the verbs *csodálkozik* ‘be surprised’ and *irigykedik* ‘be jealous’. In (12), there is no overt expletive and the absence of covert expletives is shown by the absence of agreement on the matrix verbs.⁵ Thus, the optionally occurring embedded clause is not an argument of these matrix verbs; the *because*-clause interpretation of this clause is also adjunct-like. But even if the embedded clause is not an argument to the matrix verbs in (12), the past-under-past configuration has a simultaneous

reading i.e. it denotes some cause occurring at the time. By the logic of what we have seen so far, this follows from the non-speech-reporting nature of the embedded clause.

- (12) {*Csodálkoz-t-ál* *irigyked-t-él*}, (*hogy Mari szomorú*
be.surprised-PST-2SG.INDF be.jealous-PST-2SG.INDF that Mari sad
vol-t-Ø).
be-PST-3SG.INDF
‘You were {surprised/ jealous} because of the following: “Mary is sad”.’
(apparent backshifted reading discussed in Section 2.2)

Before concluding the section, I will briefly turn to multiple embedding. In example (13), the matrix clause’s verb is the past *ámodta* ‘dreamed’, the intermediate clause’s verb is the past *látta* ‘saw’, and the deepest clause’s verb is the past *volt* ‘was’. The simultaneous reading is available between the matrix and the intermediate clause as well as between the intermediate and the deepest clause: The subject Peti dreams about Mari seeing something at the time, and what Mari sees at the time is that Zsuzsi is sad at the time.

- (13) *Peti az-t álmod-t-a*, [*hogy Mari a saját szem-é-vel*
Peti it-ACC dream-PST-3SG.DEF that Mari the own eye-3SG.POSS-INS
lát-t-a (*az-t*), [*hogy Zsuzsi szomorú vol-t-Ø*]].
see-PST-3SG.DEF it-ACC that Zsuzsi sad be-PST-3SG.INDF
‘Peti dreamed the following: “Mary sees the following: “Zsuzsi is sad””.’
(apparent backshifted reading discussed in Section 2.2)

In sum, whether the embedded clauses discussed in this section were object-clauses, subject-clauses, or non-argument clauses with a *because*-interpretation, they all had a simultaneous reading in past-under-past configurations. Essentially, all these clauses were non-speech-reporting. As we will see in the next section, this contrasts with the behaviour of speech-reporting clauses, which have only a *backshifted* reading in past-under-past configurations.

2.2 Speech-reporting clauses only have a backshifted reading

If the embedded clause describes the content of some speech act (i.e. it is *speech-reporting*), the simultaneous reading is not available in a past-under-past configuration; the embedded clause's past must be *backshifted* relative to the matrix past tense. In other words, while non-speech reporting clauses patterned together with past-under-past configurations in languages like English, this set patterns together with past-under-past configurations in languages like Japanese (Ogihara and Sharvit 2012).

Consider the complement clauses of the manner of speech verbs *mond* 'say', *rikolt* 'shout', and *morog* 'growl' (14). That these are syntactically complement clauses is indicated by the presence of the accusative-marked clausal expletive *az-t* 'it' and the definite agreement on the matrix verbs triggered by the expletive. The form of the embedded clauses is kept the same as in the previous section except that the "content of speech act" is facilitated by the describe adverb *sajnos* 'unfortunately'.

Similar to the previous section, the *hogy*-clause containing the past-tense *volt* 'was' is embedded under the past verbs *mondta* 'said', *rikoltotta* 'shouted', and *morogta* 'growled' in (14). However, unlike in the previous section, no *simultaneous* reading is available in these sentences: Peti needs to speak/shout/growl about an event that occurs before "his now". This past-in-the-past reading will be referred to as the *backshifted* reading.

- (14) *Peti az-t {mond-t-a/ rikolt-ott-a/ morog-t-a}, [hogy*
Peti it-ACC say-PST-3SG.DEF shout-PST-3SG.DEF growl-PST-3SG.DEF that
(sajnos) Mari szomorú vol-t-Ø].
 unfortunately Mari sad be-PST-3SG.INDF
 'Peti {said/shouted/growled} the following: "(Unfortunately,) Mary was sad".
 # 'Peti {said/shouted/growled} the following: "(Unfortunately,) Mary is sad".

Similar to non-speech reporting clauses, speech reporting clauses can occur as non-argument clauses, too. For example, consider sentences (15) with the intransitive verbs *üzen* 'message' and *telefonál* 'make a phone call'.⁶ Although these verbs are intransitive,

they are manner of speech verbs, and the clause occur next to them expresses the content of the message or phone call (i.e. “your message said...” and “While calling me, you said...”). Correspondingly, the past-under-past configuration in (15) only has a backshifted reading.

- (15) {*Üzen-t-él/ telefonál-t-ál*}, (*hogy (sajnos) Mari szomorú*
 message-PST-3SG.INDF/ call-PST-3SG.INDF that unfortunately Mari sad
vol-t-Ø).
 be-PST-3SG.INDF
 ‘You {messed/ called on the phone} (saying: “(Unfortunately) Mary was sad”.’
 # ‘You {messed/ called on the phone} (saying: “(Unfortunately) Mary is sad”).’

The existence of non-argument clauses belonging to both the speech-reporting class and to the non-speech-reporting class is better handled by an approach that says that clauses are internally speech-reporting or non-speech-reporting, and the semantic-pragmatic contexts filter out one class or the other. This attitude contrasts with the attitude focusing on syntactic selector of clauses (e.g. “clause X has the interpretation Y because it is the complement of the verb Z”). The latter approach starts to have problems when non-argument clauses without a clear syntactic selector come into the picture.

This internal-structure-based distinction is further supported by the observation that certain matrix verbs may co-occur with more than one type of embedded clauses. For example, while *hall* ‘hear’ in (8) encoded the direct perception of some event, (16) encodes that Zsuzsi heard Peti’s verbal report of Mary’s sadness. The presence of the speech-reporting component correlates with the availability of the simultaneous reading: While (8) referring to direct-perception had a simultaneous reading for past-under-past, (16) referring to hearing someone’s verbal report does not, i.e. Peti must report Mary’s sadness at some time preceding his “now”.

- (16) *Zsuzsi Peti-től az-t hall-ott-a,* [*hogy (sajnos) Mari {szomorú*
 Zsuzsi Peti-ABL it-ACC hear-PST-3SG.DEF that unfortunately Mari sad
vol-t-Ø/ sír-t}].
 be-PST-3SG.INDF cry-PST-3SG.INDF
 ‘Zsuzsi heard from Peti: ‘(Unfortunately,) Mary {was sad/ cried}.’
 # ‘Zsuzsi heard from Peti: ‘(Unfortunately,) Mary {is sad/ cries}.’

Certain examples suggest that the behavior of speech-reporting clauses may be generalized to the set of clauses encoding the *content of signs*. In (17), the matrix subject does not see Mary’s sadness directly but sees the signs of the sadness: the handkerchiefs left behind. In this case, the past-under-past configurations can clearly correspond to the backshifted scenario: Mary could be sad at some earlier point even if there is no speech reporting involved in this case. In general, direct evidence can only come from simultaneous perception, and past and future events can only be seen via mediators such as reports, signs, various types of evidence, video recordings etc.

- (17) *Peti az telesír-t zsebkendők-ből lát-t-a (az-t), [hogy*
 Peti the cry.in-PTCP handkerchief-PL-EL see-PST-3SG.DEF it-ACC that
(korá-bb-an) Mari szomorú volt-t-Ø].
 early-COMP-ADV Mari sad be-PST-3SG.INDF
 ‘From the used handkerchiefs, Peti saw/ inferred/ realized: “Mary was sad earlier.”’

This raises the question whether all instances of backshifted readings arise with contents of signs, and all instances of simultaneous readings arise involve direct perception of some kind. The contrast becomes harder to grasp once we get to thoughts/beliefs about the present events (which can be directly perceived) vs. thoughts/beliefs about the past (which must be recalled from the memory). Here I do not intend to provide a definite answer to this question. For convenience, I will keep using the non-speech-reporting vs. speech-reporting division, and I will use the following generalization: Non-speech-reporting clauses (in general, non-content-encoding clauses) only have a simultaneous reading and all the backshifted clauses including (17) will be structurally handled as speech-reporting clauses (in general, content-encoding clauses).

Speech-reporting clauses containing past-verbs in multiple embedding contexts show the same behaviour as biclausal structures. In (18), the matrix clause contains the past verb *rikoltotta* ‘shouted’, the intermediate clause contains the past verb *morogta* ‘growled’, and the deepest clause contains the past verb *volt* ‘was’. As we expect, the intermediate

clause must be backshifted relative to Peti’s shouting time, the deepest clause must be backshifted relative to Mary’s growling time. That is, Zsuzsi’s sadness is backshifted twice relative to Peti’s shouting time.

- (18) *Peti az-t rikolt-ott-a,* [*hogy Mari az-t morog-t-a,* [*hogy*
 Peti it-ACC shout-PST-3SG.DEF that Mari it-ACC growl-PST-3SG.DEF that
Zsuzsi szomorú volt-t-Ø]].
 Zsuzsi sad be-PST-3SG.INDF
 ‘Peti shouted saying: “Mary {growled/ #growls} saying: “Zsuzsi {was/ #} sad”’

In sum, speech-reporting clauses in past-under-past configurations have only a backshifted readings. As it will be further discussed in Sections 4, for the sake of this paper, I assume that *all* backshifted readings arise from structures patterning with speech-reporting clauses.

2.3 Multiple embedding with mixed types

While multiple-embedding examples like (13) and (18) involved two embedded clauses of the same type, two non-speech-reporting and two speech-reporting clauses, respectively, the combination of different clause-types has not been discussed. In example (19), the matrix verb is the past-tense *rikoltotta* ‘shouted’, whose complement clause is the speech-reporting clause with the verb *látta* ‘dreamed’. In turn, *látta* ‘saw’ occurs with a non-speech-reporting clause, which contains the past-tense verb *volt* ‘was’. Between the matrix and the intermediate clause, there is a backshifting: Peti shouts about some past event relative to his “now”. However, the intermediate and the deepest clause have simultaneous reading: (According to Peti), Mary perceived Zsuzsi’s sadness while it happened. This is what we expect if we consider the two stages separately: The higher, speech-reporting clause is backshifted relative to its matrix clause but the lower non-speech-reporting clause has a simultaneous reading relative to its matrix clause (the intermediate clause).

- (19) *Peti az-t rikolt-ott-a, [hogy Mari a saját szem-é-vel*
 Peti it-ACC shout-PST-3SG.DEF that Mari the own eye-3SG.POSS-INS
lát-t-a (az-t), [hogy Zsuzsi szomorú vol-t-Ø]].
 see-PST-3SG.DEF it-ACC that Zsuzsi sad be-PST-3SG.INDF
 ‘Peti shouted saying: “Mary sees the following: “Zsuzsi is sad””.’

Now consider the opposite configuration in (20), a speech-reporting clause embedded in a non-speech reporting clause. (The matrix and the intermediate clauses in (20) are switched relative to (19)). In (20), the matrix subject Mari saw the intermediate subject Peti’s shouting at her “now”. The intermediate clause’s subject Peti shouts about some past event relative to his “now”. Again, the individual stages behave in the expected way: The intermediate clause is non-speech-reporting so it has a simultaneous reading, whereas the deepest clause is speech-reporting so it has a backshifted reading. In other words, only structurally adjacent clauses interact: Just because the deepest clause is contained in the non-speech-reporting intermediate clause, the simultaneous reading is not available for the deepest clause because it is speech-reporting. In general, all what matters for the availability of the simultaneous reading of a past tense is the clause type of the next lowest clause. If a past tense must be backshifted because it occurs in a speech-reporting clause, it remains backshifted for ever.

- (20) *Mari a saját szem-é-vel lát-t-a (az-t), [hogy Peti az-t*
 Mari the own eye-3SG.POSS-INS see-PST-3SG.DEF it-ACC that Peti it-ACC
rikolt-t-a, [hogy Zsuzsi szomorú vol-t-Ø]].
 shout-PST-3SG.DEF that Zsuzsi sad be-PST-3SG.INDF
 ‘Mary saw with her own eyes: “Peti shouts the following: “Zsuzsi was sad””.’

The observation the presence of simultaneous reading solely depends on the local environment goes back to Ogihara (1996). The English example in (21) exhibits a past-under-*would*-under-past configuration (example from Sharvit 2020). Under this configuration, John cannot prepare to tell his mother “I am unhappy” even if English allows past-under-past configurations to have a simultaneous readings as in (22). Again, the problem is that

English allows the simultaneous reading for past tenses iff the next lowest tense is past. Since this is not the case in (21), the simultaneous reading for the deepest clause cannot arise even if there is a past tense two clauses higher.

(21) *A week ago, John decided that in ten days he would tell his mother that he was unhappy.*

(22) *John told his mother that he was unhappy.*

'John told his mother: "I am unhappy."' (backshifted reading set aside)

2.4 Interim summary

In this section, I demonstrated that the availability of the simultaneous reading in Hungarian past-under-past configurations depends on the clause type. Non-speech-reporting clauses in past-under-past configurations had a simultaneous reading, whereas speech-reporting clauses in past-under-past configurations had a backshifted reading. As discussed in the next section, Hungarian non-speech-reporting clauses pattern like English embedded clauses in this respect, whereas Hungarian speech-reporting clauses pattern like Japanese embedded clauses. Furthermore, multiple embedding examples showed that the presence of the simultaneous reading depends on the type of the lowest clause containing the past-tense verb, whereas higher clause-types do not matter as Ogihara (1996) observed for other languages.⁷

3 Sequence of Tense analyses

In (23), the past-verb *was* is embedded under another past verb *said*, that is, (23) is an example of past-under-past configuration in English. *Sequence of Tense* (SOT) analyses (among others, Ogihara 1996) provide a particular way to account for the *simultaneous* vs. *backshifted* readings of such past-under-past configurations (here I set aside non-SOT analyses; for a survey on the various approaches, see Sharvit 2020).

- (23) *John said Mary was pregnant.*
 Backshifted reading: ‘John said: “Mary was pregnant.”’
 Simultaneous reading: ‘John said: “Mary is pregnant.”’

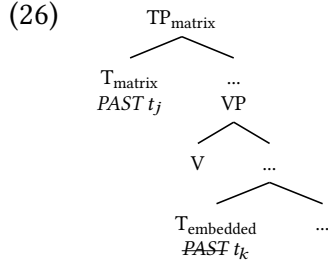
In the backshifted reading, which was associated with Hungarian speech-reporting clauses in Section 2.2, John makes a claim about Mary’s pregnancy preceding “his now”. The formula in (24) is the LF of the backshifted reading of (23). In this paper, $[PAST t_k]$ complexes are handled as quantifiers with the meaning “some time prior to t_k ”, which need to be raised to the edge of their respective clauses. Within the T heads of the clauses, quantifier raising leaves behind t_j variables, which are of the type i , a special type for times. Hence, the quantifier $[PAST t_0]$ (i.e. “some time preceding our now”) in (24) occurs at the edge of the matrix clause and binds John’s speaking time in $[say t_1]$. Furthermore, John’s speaking time $[say t_1]$ binds the variable in $[PAST t_2]$ (t_2 is “John’s now”), which is a quantifier raised to the edge of the embedded clause. That is, John’s speaking time is some past time relative to “our now”, and the pregnancy happens at some past time *backshifted* relative to “John’s now”. By and large, the pregnancy is past-in-the-past (for a more complete overview, see Sharvit 2020).

- (24) $[PAST t_0] \lambda_1 [John [say t_1] \lambda_2 [[PAST t_2] \lambda_3 [Mary [be t_3] pregnant]]]$

In the *simultaneous* reading of (23) associated with Hungarian non-speech reporting clauses, John’s makes a claim about Mary’s pregnancy at “his now”. Sequence of Tense approaches (e.g. Ogihara 1996) derive this reading via proposing the “deletion”-under-Agree rule in (25). The Sequence of Tense Rule does not have a phonologically realized effect, i.e. phonological form of the simultaneous and the backshifted reading is the same (hence the quotation mark around “deletion”).⁸

- (25) **The Sequence of Tense Rule:** If some $[PAST t_j]$ c-commands another $[PAST t_k]$, the c-commanded $[PAST t_k]$ can undergo deletion under Agree i.e. become $[PAST t_k]$ (non-indexical present).

The application of the Sequence of Tense Rule is depicted in (26): Since the $[PAST\ t_j]$ located in the matrix T head, it c-commands $[PAST\ t_k]$ located in the embedded T head, the *PAST* in the latter can undergo deletion under agreement.



I assume that the quantifier raising of $[PAST\ t_k]$ complexes and the application of the Sequence of Tense rule in (25) are ordered. The *PAST*-deletion happens as soon as the c-command relation specified in (25) holds in the *Narrow Syntax*. (Notice that the deletion still has no affect on the phonological form of the embedded past tense.) Since the c-command relation holds as soon as the matrix T head is merged in (26), the deletion precedes quantifier raising. In the LF of the simultaneous reading in (27), I assume that the $[~~PAST~~\ t_2]$ complex ceases to be a quantifier after deletion: It becomes a non-indexical present, which is nothing but a t_2 variable in the embedded T head. This later gets simply bound by the matrix $[say\ t_1]$ complex, i.e. John reports Mary's pregnancy occurring at "his now".

- (27) $[PAST\ t_0]\ \lambda_1\ [John\ [say\ t_1]\ \lambda_2\ [Mary\ [be\ [~~PAST~~\ t_2]]\ pregnant]]$ becomes
 $[PAST\ t_0]\ \lambda_1\ [John\ [say\ t_1]\ \lambda_2\ [Mary\ [be\ [t_2]]\ pregnant]]$

The presence of the Sequence of Tense rule in (25) has been usually handled as a language-wide parameter, e.g. since no SOT rule can *ever* apply in Japanese, no simultaneous reading is available in past-under-past configurations (Ogihara and Sharvit 2012:p. 640):

- (28) *2005-nen ni Joseph-wa [Mary-ga (#sono-toki) zibun-o asisi-tei-ta-to]*
 2005-year at Joseph-TOP Mary-TOP that-time self-ACC love-PROG-PAST-that
sinzi-tei-ta.
 believe-PROG-PAST
 ‘Joseph to himself, in 2005: “Mary liked me”.’
 # ‘Joseph to himself, in 2005: “Mary loves me now”.’

4 High attachment and Cyclic Sequence of Tense: A syntactic account

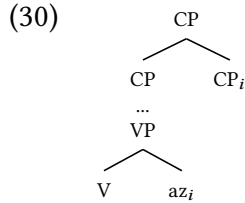
4.1 The syntax of size-dependent high attachment

É. Kiss (2002) notes that Hungarian embedded clauses cannot end up within the VP. In (29), the expletive of the clause gets sublative case. The expletive itself can be *pro*-dropped but case-remnant *rá* remains overt in the VP-internal position (Marác 1989). (In the case of accusative expletives the *pro*-drop is complete so it is the definiteness agreement on the verb shows the general presence of the expletive.) Importantly, the embedded clause (29) cannot occur within the VP, it must occur behind the VP- and TP-modifying adjuncts in the right periphery, in a position separated from the main verb and the clausal expletive.

- (29) *Mari hivatkoz-ott {ar-ra_k/ Ø-rá_k} { tegnap/ a buli-ban}, [hogy Peti*
 Mari refer-PST.3SG.INDF it-SUB pro-SUB yesterday the party-IN that Peti
*alud-t-Ø]_k { *tegnap/ *a buli-ban}.*
 sleep-PST-3SG.INDF yesterday the party-IN
 ‘Mari referred to the following in the party yesterday: “Peti slept.”’

The embedded clause’s position can be derived as either base-generation in the right-periphery or extraposition. For instance, É. Kiss (2023) relies on the high base-generation analysis in (30): The embedded CP is adjoined to the matrix CP and binds the clausal expletive *az* ‘it’ in the matrix clause. In contrast, Egressy (2023) proposes that the clause and the expletive are generated in a VP-internal small clause, and then the embedded clause gets extraposed. In this paper, I follow a modified version of É. Kiss’s (2023) analysis: Hungarian argument and non-argument clauses introduced by *hogy* ‘that’ are

base generated in a high-adjoined position. Notice that *adjunction* refers to the mode of combination rather than the thematic status or argumenthood. Argument clauses bind expletives in the matrix clause’s argument structure, whereas adjunct clauses do not.⁹



I modify É. Kiss’s analysis and assume that Hungarian *hogy*-clauses are not uniform in size, they are not all just CPs. More specifically, I assume that Hungarian has no CP layer in its clause structure, and in syntactic terms, *hogy* ‘that’ is not a C head but merely an *edge marker* (a term by Manetta 2006, 2011) adjoined to the highest projection of the embedded clause. One piece of evidence for this is shown in (31): *hogy* does not occur at the edge of an embedded clause (i.e. it is not involved in syntactic subordination), but next to an echo question. Furthermore, *hogy* is capable of occurring next to subjunctives, tensed clauses, embedded questions, or even infinitivals (Koopman and Szabolcsi 2000; Kenesei and Szeteli 2022). In short, *hogy* is not restricted to clausal embedding and does not seem to be sensitive to the category of the embedded clause in a way as English *that* can only occur next to embedded tensed declarative clauses. For a more detailed discussion, see Egressy (2023; 2025).

- (31) A: Az Ötös Lottó heti várható főnyereménye 500 millió forint.
the five lottery weekly expected jackpot 500 million HUF
B: *Hogy mennyi?!*
that how.much
A: ‘This week the estimated jackpot of Lottery 5 is 500 HUF million.’ B: ‘How much?!’

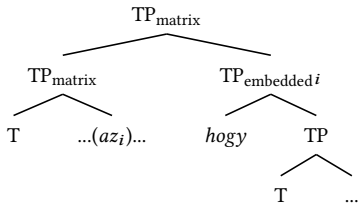
Since embedded clauses in Hungarian lack the CP layer, tensed *hogy*-clauses will be handled as bare TPs by default. For the attachment-site of embedded clauses, I adopt a cyclic version of Williams’ Level Embedding Conjecture (Williams 2003; Poole 2023;

Meadows 2023) in (32). According to the original Level Embedding Conjecture, an αP can only be embedded in a matrix clause that is also built up to αP , before the embedding of αP a place holder occurs in the structure. The original version brings about a number of countercyclic derivations: While the matrix α merges in, we need to go back and modify the VP by the insertion of an αP , e.g. an embedded TP clause can only be embedded in the matrix VP when the matrix T merges in. The cyclic version in (32) avoids this problem by relying on high-attachment, whereas expletives occurring in the argument structure of the matrix verb may be handled as the overt realizations of the place-holder.

- (32) **Cyclic Level Embedding Conjecture:** An αP can only adjoin to a structure that is also built up to αP .

The size-dependent high-attachment analysis of Hungarian embedding is depicted in (33). The key two assumptions are the following: i) *hogy* is not a C head but an adjunct to the highest projection of the embedded clause, which is by default a TP; ii) the embedded clause is adjoined to an equal-sized matrix projection in accordance with (32). (Again, az_i ‘it’ only participates in the embedding of argument clauses.)

- (33) Non-speech-reporting embedded clauses

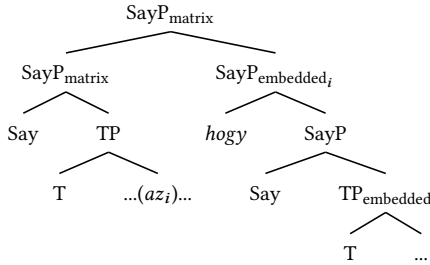


However, not all embedded clauses are TPs; speech-reporting clauses involve an extra operator heading an layer in the structure (see Kratzer 2006; Moulton 2009; Deal 2020; Krifka 2023). Syntactic layers encoding “content” or “saying” have been shown to receive an overt realization in languages like Uyghur (Major 2021) or Buryat (Bondarenko 2022). In a similar way, Deal (2020) proposes a size-difference between the complements of speech verbs and non-speech verbs to account for differences in indexical shift. While

Krifka (2023) breaks down this layer into CommitmentP, JudgementP, and ActP, other authors like Bondarenko (2022) use a single Cont(ent)P and Major (2021) uses a single SayP projection for the additional layer of speech-reporting clauses. Although I discussed in Section 2.3 that the behavior of speech-reporting may be generalized to the class of embedded clauses encoding the content of some (not necessarily verbal) signs, the ContP label would raise the question why dreams and thoughts are not encoded in ContPs.¹⁰ Therefore, I will be more careful and use SayP for *speech-reporting* clauses with the note that SayP can also describe the content of non-verbal signs. I will continue to assume that every clause that has a backshifted reading is speech-reporting (or sign-reporting), i.e. a SayP, and every clause that has a simultaneous reading is non-speech-reporting (non-speech-reporting), i.e. a TP.

By the Cyclic Level Embedding Conjecture in (32), embedded SayPs are attached one step higher, at matrix SayP as in (34).

(34) Speech-reporting embedded clauses



The syntactic evidence for the presence of this extra projection comes from the high position of sentential evaluative adverbs like *sajnos* ‘unfortunately’ in (35) (e.g. Egedi 2021), which modify the SayP layer of speech-reporting clauses. Such adverbs completely unavailable in the clause of *álmodik* ‘dream’, which requires a TP-sized clause. Moreover, with ‘unfortunately’ occurs in their complement clauses, *hall* ‘hear’ and *gondol* ‘think’ need to mean ‘hear a report of’ and ‘tell to himself’, respectively, i.e. these clauses are speech-reporting SayPs even if these verbs would be compatible with TPs, too. Notice that only the backshifted reading is available for the grammatical/felicitious sentences in (35).

- (35) *Pál azt {mond-t-a/ hall-ott-a/ gondol-t-a/ #álmod-t-a},*
 Pál it.ACC say-PST-3SG.DEF hear-PST-3SG.DEF think-PST-3SG.DEF dream-PST-3SG.DEF
 [_{SayP} *hogy sajnos* [_{TP} *Ili szomorú volt*]].
 that unofrutantly Ili sad be-PST.3SG.INDF
 ‘Pál {said/ heard someone saying/ told to himself}: “Unfortunately, Ili {is/ #was} sad.”’

Finally, even if *hall* was compatible with both TPs as in (8) and SayPs as in (16), once the matrix subject is not capable of processing and evaluating the content of (verbal) signs, such evaluative adverbs limited to speech-reporting clauses are unavailable. As expected, the embedded TP in (36) has a simultaneous reading.

- (36) *A mikrofon hall-ott-a, [_{TP}hogy (??sajnos) közeled-t-ünk].*
 the microphone hear-PST-3SG.DEF that unfortunately approach-PST-3PL.INDF
 ‘The microphone heard: “(??Unfortunately) they are approaching.”’

In sum, while clear-cut instances of non-speech reporting clauses are TPs and adjoined to matrix TPs as in (33), speech-reporting clauses are all SayPs and adjoined to matrix SayPs as in (34) (for a similar symmetrical adjunction for adverbial clauses, see Endo and Haegeman 2019). The difference in the attachment heights for embedded clauses of different sizes has benefits beyond this paper: It explains Williams Cycle effects in Hungarian as matrix movement and agreement probes can only “look into” if they are high enough in the matrix structure. For further discussion on Williams Cycle effects in Hungarian, see Egressy (2023, 2025).

4.2 Cyclic Sequence of Tense

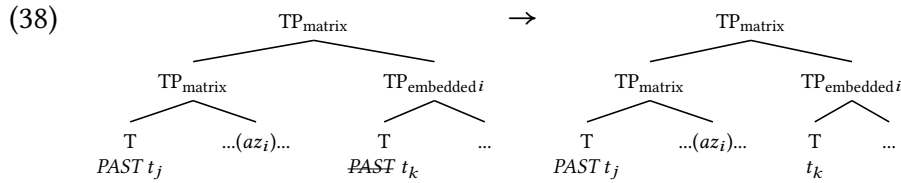
Once the size-dependent attachment of Hungarian embedded clauses has been established, we only need two adjustments to the Sequence of Tense Rule in (25) in order to account for the data presented in Section 2. The new rule, the Cyclic Sequence of Tense (henceforth, CSOT) is defined in (37). On the one hand, the rule in (37) operates in the spirit of Cyclic Agree (Rezac 2003; Béjar and Rezac 2009): The search space for deletion is not limited to the complement of the T head bearing the *PAST* feature but the “probing” can continue

as long as the T projects (in this case, the higher *PAST* is the probe and the lower *PAST* is the goal).¹¹ Notice that “vertically embedded structures” like the one in (27) remain compatible with this deletion rule because the matrix TP contains the embedded TP. On the other hand, while the original rule in (25) applied optionally, I assume that the Cyclic Sequence of Tense rule applies as soon as its structural specification is met. That is, this rule does not apply to complete trees at LF but applies whenever it can in narrow syntax. Consequently, in order block it, we need to change the structural configuration.

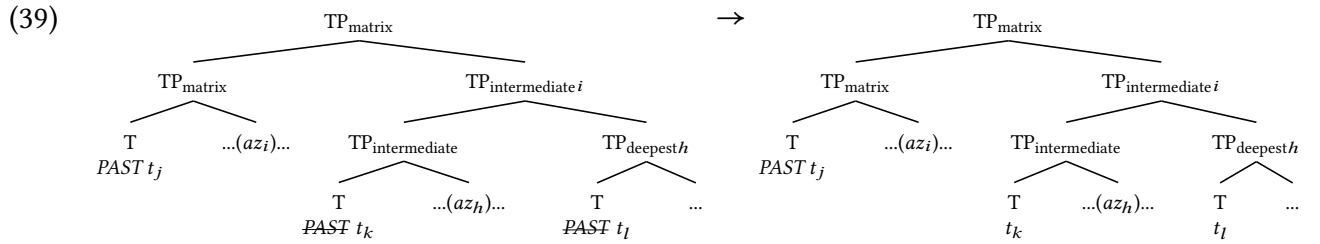
(37) **The Cyclic Sequence of Tense Rule (CSOT):**

At any point in the derivation, if a projection of $[PAST\ t_j]$ contains another $[PAST\ t_k]$, the contained $[PAST\ t_k]$ must undergo deletion under agreement, i.e. become $[~~PAST~~\ t_k]$ (non-indexical present).

In the TP-TP adjunction structure of non-speech-reporting clauses like (7) depicted in (38), the CSOT in (37) can apply because the highest matrix TP, which is a projection of the matrix T head featuring *PAST*, contains the embedded *PAST* T head. The embedded deleted *PAST* becomes invisible for the LF computation, that is, all what remains in the embedded T head is the t_k variable, i.e. the non-indexical present as represented in the right. That is, the past-under-past in TP-TP adjunction structures after the obligatory CSOT-application has a *simultaneous* reading, as expected. (This is why the past-under-past and present-under-past settings are synonymous for non-speech-reporting clauses, as mentioned in Footnote 7.) Since CSOT applies as soon as the containment relation holds, no quantifier raising is needed at LF. The options for the semantics of (38) are discussed in the Section 5.

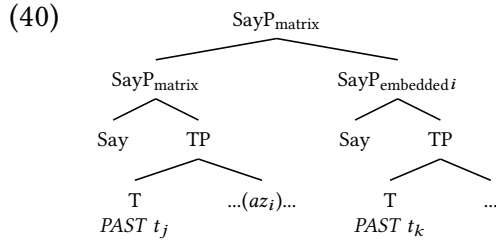


In the multi-layered embedding of non-speech-reporting clauses like (13), the pattern is the same: Every embedded TP adjoins to its local matrix clause as in (39). Since the deepest *PAST* is contained in the highest TP projection of the intermediate clause, which is a projection of the intermediate *PAST*, so the deepest *PAST* is deleted. Furthermore, the intermediate *PAST* also gets deleted, once the intermediate TP adjoins to the matrix TP because the highest matrix TP is a projection of the matrix *PAST*. In short, the simultaneous reading is derived in both cases.



Notice that approaches that assume that the deletion rule applies to complete trees at LF needs to impose extrinsic ordering to avoid the deletion of the intermediate *PAST* before the lowest *PAST* in (39). Without this ordering, we would need to allow the deletion of the deepest *PAST* by the matrix *PAST*, which seems to be problematic independently (see Ogihara’s generalization in Section 2.3 and below). In contrast, since CSOT applies whenever the containment relation holds in the bottom up-derivation of Narrow Syntax (i.e. the intermediate and the deepest clauses merge before the merger of the intermediate and the matrix clauses), the right ordering does not need to be stated separately in my analysis.

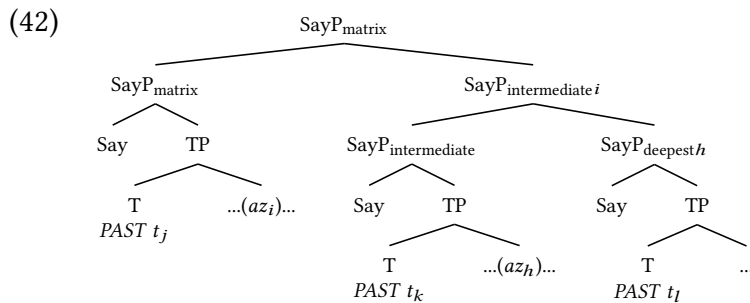
As it is depicted in (40), speech-reporting clauses like the one in (14) are SayPs and they adjoin to matrix SayPs by the Cyclic Level Embedding conjecture in (32). In (40), the CSOT in (37) cannot apply because the embedded SayP attaches too high to be contained in the matrix TP. Consequently, the embedded *PAST* stays in place for LF, that is, the embedded clause is *backshifted* relative to the matrix clause, as expected.



The possibility of *PAST*-deletion in smaller TPs and the impossibility of *PAST*-deletion in larger SayPs can be described as a Williams Cycle effect (Williams 2003, 2013; Müller 2014a,b; Poole 2023; Meadows 2023; Egressy 2023, 2025) in Agree in (41), which is comparable to the size-sensitivity of Hindi Long Distance Agree (Keine 2016, 2019, 2020).

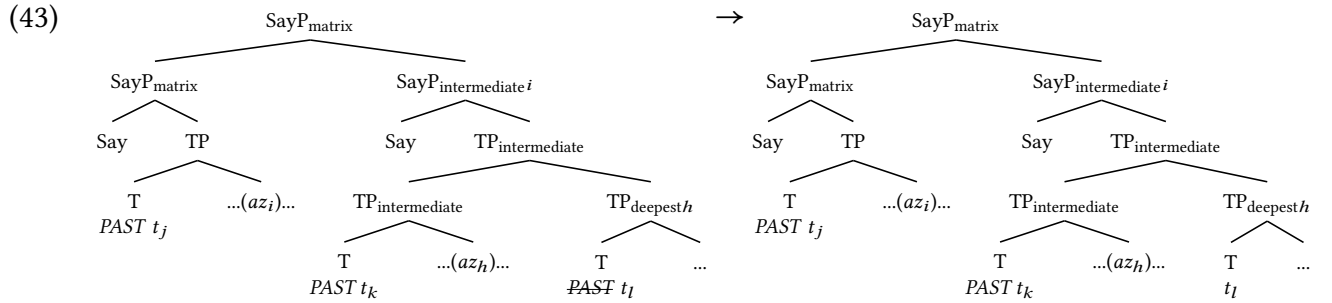
- (41) **Williams Cycle in Agree:** No Agree from β can search in αP , if $\alpha > \beta$ in the functional sequence.

The multi-layered embedding of speech-reporting clauses like the one in (18) works in the same way. In (42), since the deepest past is not contained in either the intermediate TP or the matrix TP, and the intermediate *PAST* is not contained in the matrix TP, the CSOT rule cannot apply. Therefore, the deepest *PAST* backshifts the deepest clause relative to the intermediate clause and the intermediate *PAST* backshifts the intermediate clause relative to the matrix clause.

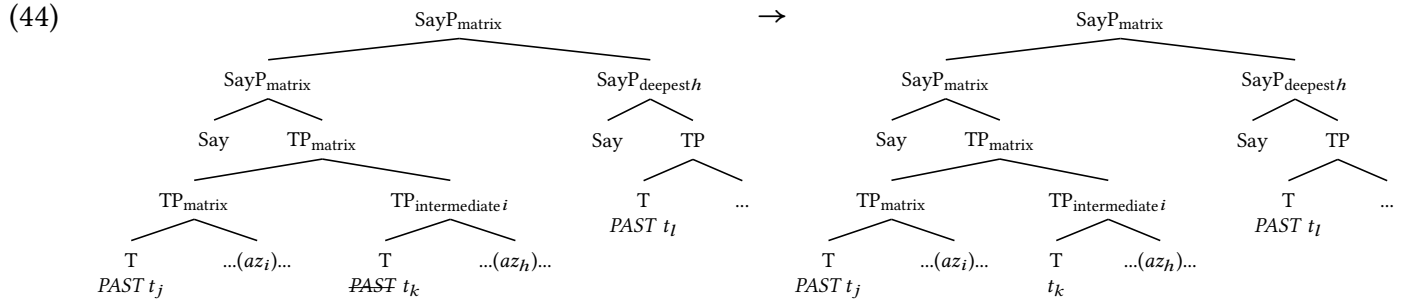


As it was discussed in Section 2.3, non-speech-reporting clauses can be embedded under speech-reporting ones and vice versa. The simpler case is when the deepest clause is a non-speech reporting clause as in (19). Such configurations are depicted in (43): While the deepest TP adjoins to the intermediate clause's TP projection, the intermediate SayP

adjoins to the matrix clause's SayP projection. Since the deepest *PAST* is contained in the highest intermediate TP, the CSOT-deletion of the deepest *PAST* is possible. Therefore, the deepest *PAST* becomes a non-indexical present, i.e. the deepest clause is simultaneous with the intermediate clause, as expected. However, the intermediate *PAST* is not contained in the matrix TP because the intermediate SayP attaches too high. Consequently, the intermediate clause is backshifted relative to the matrix clause, as expected.



The embedding of a larger speech-reporting SayP under a smaller non-speech-reporting TP as in (20) is structurally more complex because by the Cyclic Level Embedding Conjecture in (32), a smaller structure contain a larger one. In the tree structure in (44), the intermediate TP adjoins to the matrix TP. Since the highest matrix TP contains the intermediate *PAST*, the CSOT-deletion happens, and the intermediate T head becomes a non-indexical present. Therefore, the intermediate clause is simultaneous with the matrix clause, as expected. Since the intermediate clause does not have a SayP projection to which the deepest SayP could adjoin, the deepest SayP adjoins to the matrix SayP. Since the deepest *PAST* is contained in neither the matrix nor the intermediate TP, the deepest *PAST* cannot undergo CSOT-deletion. Hence, the deepest clause remains backshifted relative to the matrix clause, which means the same as backshifting relative to the intermediate clause because the matrix and the intermediate clause are simultaneous. This derives the expected reading following Ogihara's generalization discussed in Section 2.3: The CSOT-deletion of the deepest *PAST* is impossible from the matrix clause.



5 Conclusion and open questions

In this paper, I demonstrated that in Hungarian, non-speech-reporting embedded clauses (TPs) allow a *simultaneous* reading for the past-under-past configuration, whereas speech-reporting embedded clauses (SayPs) only have a *backshifted* reading. The possibility of *PAST*-deletion was described as a Williams Cycle effect (e.g. Williams 2003) in Agree (e.g. Keine 2016). To analyze this, I assumed that all embedded clauses adjoin high, when the matrix clause reaches their size: TPs to TPs and SayPs to SayPs. I assumed furthermore that the Cyclic Sequence of Tense rule can apply as long as a projection of the matrix *PAST* contains the embedded *PAST*. This rule could apply to TP-TP adjunction structures but not to SayP-SayP adjunction structures, thus the embedded *PAST* could only be deleted in the former case. Ogihara's generalization was also derived from the idea that SayP-sized embedded clauses can never adjoin low enough to be affected by the CSOT rule. This strongly syntactic analysis is capable of handling the intra-linguistic variation of Hungarian past-under-past configurations, whereas previous accounts handled the availability of simultaneous reading as a language-wide parameter.

Space did not allow me to provide a detailed semantics for Hungarian in the main text; here I see two options. The simpler option is to say that high-attached clauses are capable of reconstructing into the VP-internal position of their expletives, i.e. the LFs of Hungarian sentences does not differ from the LFs of English and Japanese sentences in

Section 3. The challenges for this option are that adjunct clauses do not have expletives (at least not overt ones) and mainstream generative syntax assumes that reconstruction is only possible into a positions where clauses have a syntactic trace (rather than some pronoun). Therefore, I sketch an alternative semantic analysis in the appendix which keeps the high-attachment configurations from syntax and handles argument and adjunct clauses alike. This alternative is more in line with the spirit of this paper.

Obviously, it remains an open question whether this analysis can be extended to SOV (and S-CP-V) languages like Japanese, which pose some challenge to the high-attachment analysis. Furthermore, this paper did not discuss any kind of adjunct clauses not introduced by *hogy* ‘that’, and relative clauses were also ignored. Finally, the exact source of different behavior of the two classes of clauses with respect to Sequence of Tenses remains to be explored. According to É. Kiss (2023), Hungarian finite embedding emerged from the gradual integration of paratatically embedded finite clauses into the matrix clause as the language developed from SOV to SVO. Possibly, just as the indexical shift of embedded pronominals gradually disappeared along this SOV-SVO shift in Old Hungarian and Middle Hungarian (É. Kiss 2023), the simultaneous reading is first available only for a subset of Modern Hungarian clauses.

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Notes

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second one in Spring, 2023. The surveys had 15 and 35 native-speaker participants, respectively.

² The glossing follows the Leipzig Glossing rules. The abbreviations are the following: 1 = first person, 2 = second person, 3 = third person, ABL = ablative, ACC = accusative, ADV = adverb, COMP = comparative, DEF = definite, EL = elative, IN = inessive, INDF = indefinite, INF = infinitive, INS = instrumental, PL = plural, POSS = possessive, PRS = present, PST = past, PTCP = participial, SG = singular, SUB = sublativ, SUBJ = subjunctive

The uncited Hungarian data comes from my own data collection. Although I am a native speaker of Hungarian, the grammaticality of critical sentences were double-checked with native-speaker linguists and in a survey with 15 native speakers (conducted Summer, 2022). The Hungarian data of the paper comes from the standard dialect spoken in Budapest.

³ While the embedded clauses in (7) and (8) were factive, this does not have to be the case for verbs of direct perception. The pronominal *úgy* explicitly introduces the possibility of wrong perception in (45), while the temporal structure of the clause remains the same as in (7). Since factivity does not seem to correlate with the temporal properties of embedded clauses, I will not discuss it here. (At the same time, a partial correlation may be is that the embedded clauses of manner-of-speech verbs cannot be factive, and they only have a backshifted reading.) With respect to factivity/veridicality in Hungarian, I follow Abrusán (2025), who by-and-large reduces factivity/veridicality to the *position* of embedded clauses in the matrix clause. In contrast, the SOT-properties are derived here from the internal *size* of embedded clauses. See Egressy (2023) for more discussion on the size vs. position distinction.

- (45) *Peti a saját szem-é-vel úgy lát-t-a (az-t), [hogy Mari szomorú*
 Peti the own eye-3SG.POSS-INS so see-PST-3SG.DEF it-ACC that Mari sad
vol-t-Ø]. (Pedig vidám vol-t-Ø)
 be-PST-3SG.INDF but happy be-PST-3SG.INDF
 ‘Peti (potentially incorectly) saw with his own eyes: “Mary is sad.” (But she was happy.)’

⁴ É. Kiss (2002) refers to this phenomenon as *loose adjunction*. I will not use the term because in my paper, the word *adjunction* primarily refers to a mode of combination, and not argumenthood. (As we will see later, even argument clauses are syntactically *adjoined*, too.)

⁵ In Hungarian, only nominative and accusative expletives can be dropped completely, while expletives bearing other cases must have a P-remnant (Dékány and Hegedűs 2021). Nominative and accusative expletives (whether overt or covert) trigger subject and object agreement, respectively. Consequently, if clauses like (12) had expletives, we could see their presence in one way or another.

⁶ É.Kiss (2002) mentions that in her Hungarian variety, *telefonál* ‘to make a phone call’ has a transitive version. In my variety, it cannot be transtive.

⁷In this paper, I do not discuss present-under-past configurations. For most cases, the embedded present in Hungarian seems to be non-indexical, i.e. it does not access the “our now”. Therefore, (46) is felicitous with *two years ago* (i.e. Mary’s pregnancy does not need to last for two years). For the non-speech-reporting clauses above, present-under-past is the only way encode the simultaneous reading; for non-speech-reporting clauses, the present-under-past and the past-under-past configurations both encode simultaneous readings.

- (46) (*Két évvel ezelőtt*) *Peti az-t {a saját szem-é-vel lát-t-a/ rikolt-ott-a},* [
 two years ago Peti it-ACC the own eye-3SG.POSS-INS see-PST-3SG.DEF shout-PST-3SG.DEF
hogy Mari {szomorú/ terhes} Ø].
 that Mari sad pregnant be.PRS.3SG.INDF
 ‘Two years ago, Peti {saw with his own eyes/ shouted}: “Mary is {sad/pregnant}”.’

However, there seems to be an exceptional case in which Hungarian embedded present denotes the utterance time of the matrix speaker. In (47), *two years ago* is infelicitous for anyone who knows football: It is usually not the case that a player scores a goal at some point and that single goal actively keeps the team on the top of the billboard for two seasons.

- (47) (*#Két évvel ezelőtt*) *Szoboszlai elér-t-e* [*hogy a Liverpool az első*
 two years ago Szoboszlai achieve-PST-3SG.DEF that the Liverpool the first
 Ø].
 be.PRS.3SG.INDF
 Without *two years ago*: ‘Szoboszlai achieved that Liverpool is the first (right now).’
 With the *two years ago*: ‘Two years ago, Szoboszlai achieved that Liverpool is the first’
 (i.e. It’s still his two-year-old achievement that Liverpool has been the first in the last two years.)

The matrix predicate *elér* ‘achieve’ is special in that it tolerates subjunctive clauses but cannot stand with non-factive clauses, whereas most Hungarian matrix predicates cannot stand with both subjunctives and tensed clauses and their embedded clause can be de-factivized (e.g. Tóth 1999).

⁸From a morphosyntactic point of view, the idea of “cross-clausal” may occur as unusual compared to other instances of agreement in the languages. A parallel construction may be *Only I got a question I could answer*, which has a reading in which no other *x* got a question that *x* could answer. According to one analysis of this reading, the lower *I* gets deleted under agreement with the higher *I* in a way that only an *e*-type variable is left behind.

⁹One problem with the high-base generation analysis is that Hungarian embedded clauses reconstruct under the matrix subject for Condition C (Kenesei 1994). The extraposition analysis can be adjusted in a way that it remains compatible with the rest of the ideas presented, e.g. by locating the SOT rule in the LF or by the obligatory topic movement of subjects to a position c-commanding the embedded clause. For the sake of simplicity, I will keep the high base-generation account in the main text. The high base-generation

account can be maintained if we assume that although clauses are base-generated high, they reconstruct in the position of the clausal expletive.

¹⁰More specifically, while the speech-reporting class seems to homogeneously describe propositional content, the non-speech-reporting class is not homogeneous. According to Bondarenko (p. c.), while emotive non-speech reporting clauses like the complement clause of ‘think’ and ‘believe’ can be described as “the experiencer has a mental state colored by this emotion with content”, content-less emotive states like the complement clause of ‘see’ and ‘hear’ can be described as “the experiencer feels this emotion, because of the situation of this kind”. Interestingly, if we replace ‘think’ in (10) with ‘have a thought’ in (48), the simultaneous reading becomes less accessible.

- (48) *Peti-nek az vol-t a gondolat-a, [hogy Mari szomorú vol-t-Ø].*
 Peti-DAT it be-PST-3SG.INDF the thought-3SG.POSS that Mari sad be-PST-3SG.INDF
 ‘Peti had the thought: “Mary was sad.” but ?? ‘Peti had the thought: “Mary is sad.”’

¹¹My assumptions on Cyclic Agree are stricter than those of Clem (2022), because she allows maximal projections functioning as adjuncts to probe into their host phrases, whereas the definition in (37) excludes this possibility. This is essential as Clem’s version would allow the deletion of the matrix past under agreement with the embedded with the matrix PAST.

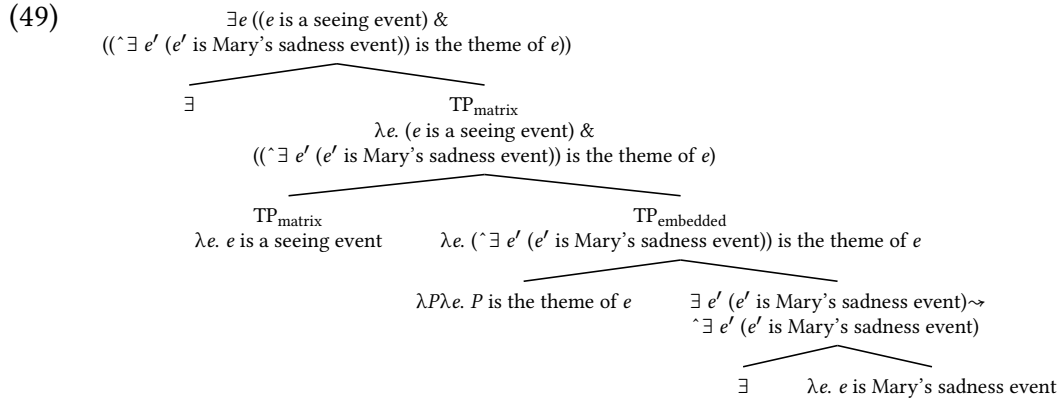
1 Appendix:

The semantics of high-attachment and (non-)deleted *PAST*s

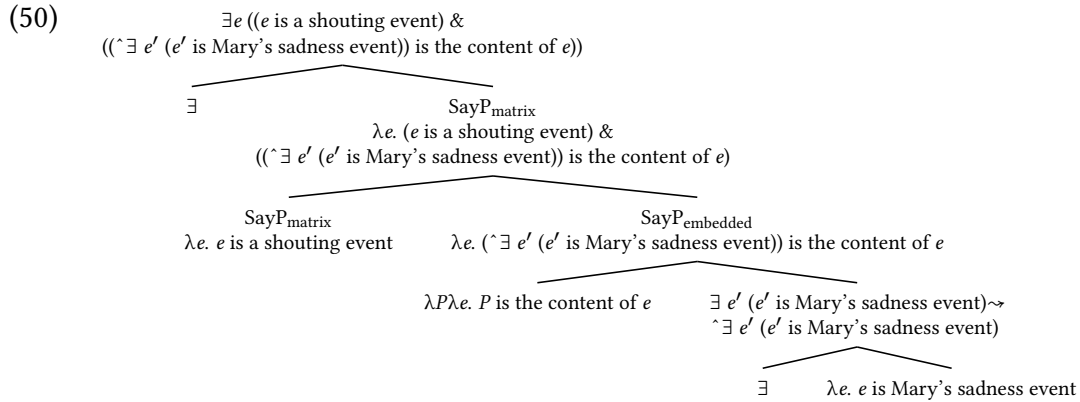
In this sketch of semantic analysis, High-adjoined *hogy*-clauses are treated in the spirit of Neo-Davidsonian event-semantics (e.g. Kratzer 2002). Section 5.1 outlines the general semantics of clause-adjunction without the discussion of times. Times will be added to the derivations in Section 5.2. My semantic derivations are given in a simplified form, I set aside details of *factivity* and *mistaken perspective* (when the matrix subject is mistaken about their “now”).

1.1 Event semantics for high attachment

In general, the matrix clause as well as the embedded clause are assumed denote sets of events. For a clause like *John saw that Mary was pregnant* (or (7) in Section 2.1), the matrix clause is a set of seeing events, whereas the embedded clause denotes a set of events whose theme is Mary’s pregnancy. The compositional semantics (for now, without the tenses) is shown in (49). The embedded clause starts out as a set of Mary’s sadness events, after this set of events is closed over existentially, and turned into an *st*-type proposition (von Stechow and Heim 2011). This merges with the operator that encodes the theta-role of the embedded clause, which is *theme* in this particular case. Above this, the matrix clause (a set of seeing events) and the embedded clause (another set of events whose theme is Mary’s sadness) can combine via Predicate Modification (i.e. we take the intersection of the two sets). After this merger, there is an existential closure over events. As mentioned above, I set aside the problems of factivity (see Footnote 3 and Abrusán 2025).



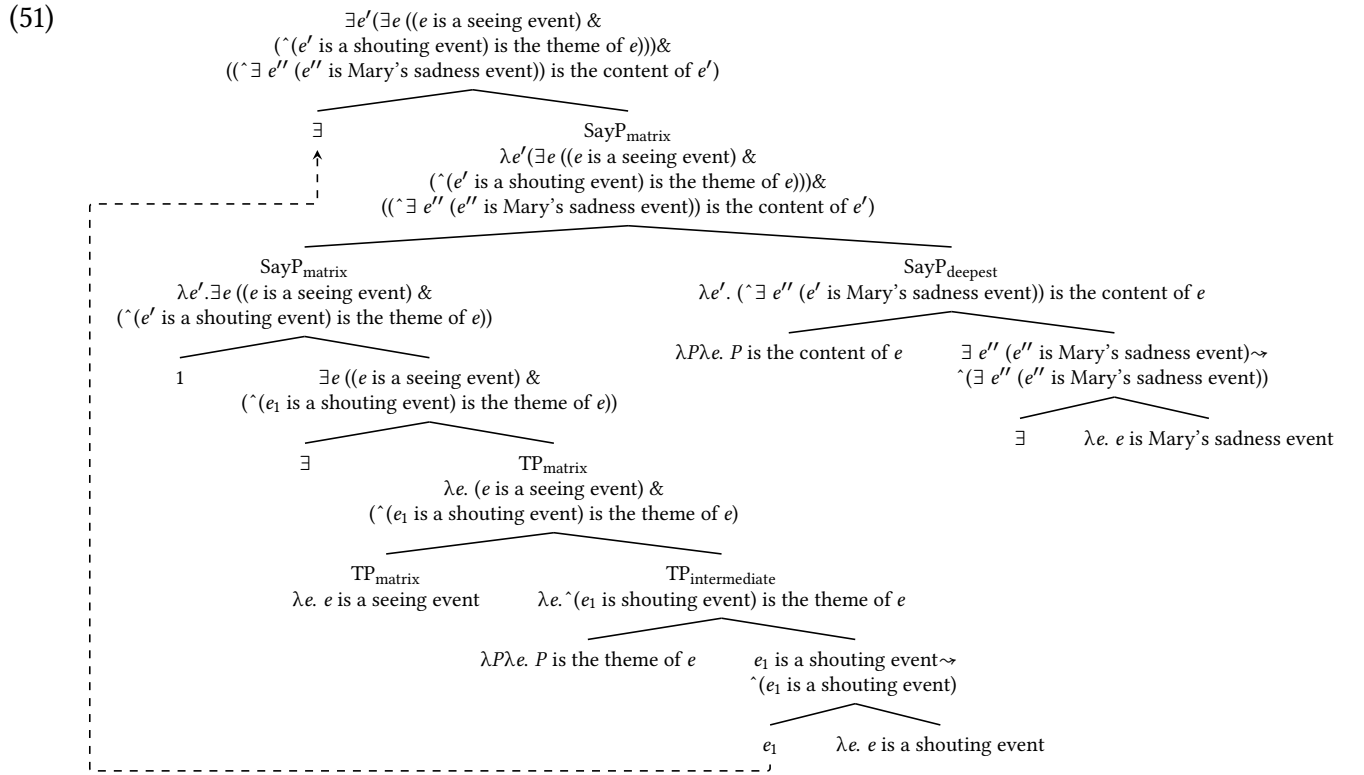
The event-semantics of speech-reporting clauses like *John shouted that Mary was sad* in (14) is only different in that the matrix clause involves another activity and the proposition of the embedded clause does not serve as the *theme* but as the *content* of the matrix clause:



All embedded clauses follow the general pattern exemplified by (49) and (50). Clauses can get further theta-roles e.g. *cause* for non-argument non-speech-reporting clauses in (12). Furthermore, verbs like *see* can stand with either non-speech-reporting clause, which has a *theme* role, or a speech-reporting clause, which has a *content* role.

Most instances of multiple embedding, specifically, the ones involving two non-speech-reporting clauses, two speech-reporting clauses, or a non-speech-reporting clause under a speech-reporting clause, follow the pattern of (49) and (50). For example, the topmost node of (49) could merge with a *theme*-operator or *content*-operator could adjoin to a set of dreaming or shouting events, respectively.

The tree in (51) depicts the slightly more special SayP-under-TP (i.e. larger-under-smaller) setup like (20) whose syntax was depicted in (44). What is special about these cases is that the deepest clause attaches to the matrix clause, to a position above the intermediate clause. In order to deal with this, the existential quantifier over events in (51) undergoes quantifier raising from the intermediate clause leaving behind the variable e_1 . After the merger of the matrix clause and the intermediate clause, and the existential closure over their intersection, there is an abstraction over the Index 1. That is, we have a set of events again. This set of events is a SayP and can merge with the deepest SayP, which denotes a set of events whose content is Mary's sadness.



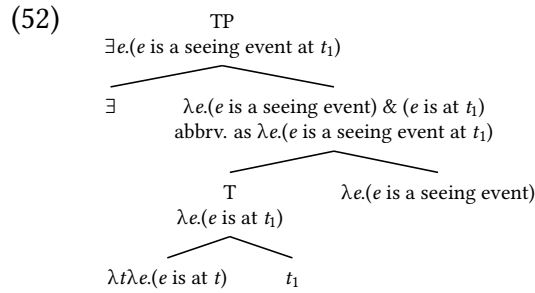
In sum, the high-adjunction of embedded clauses can be described with event-semantics. Whether the embedded clauses were subjects, objects, or non-arguments to the matrix clause, they all combined with the matrix clause via Predicate Modification. I also demonstrated that the raising of existential quantifiers over events deals with structures in which

the syntax introduced in Section 4 dictates that embedded clause and the embedded clause of the embedded clause both adjoin to the matrix clause.

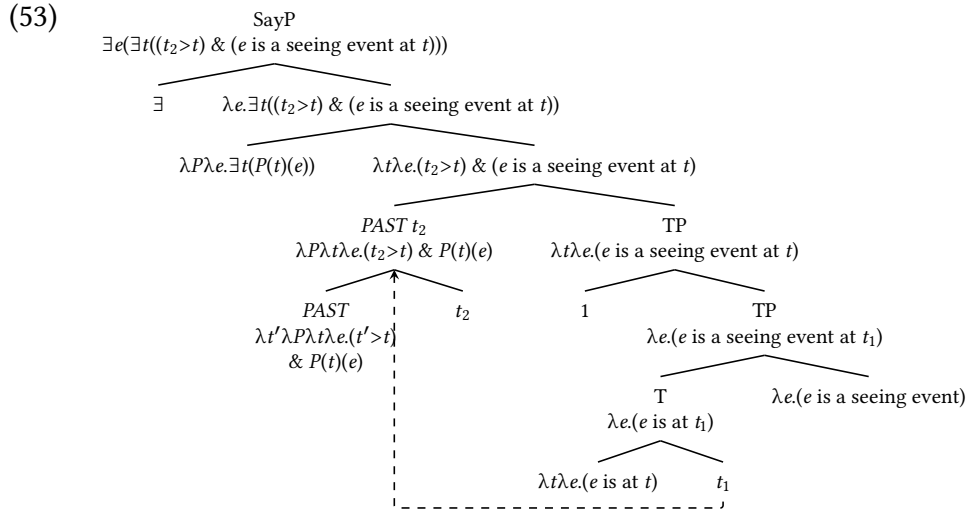
1.2 Adding tense to events

This section assigns LFs to the syntactic derivations in Section 4.2. The basis for these LFs will be the event-based derivations in Section 5.1.

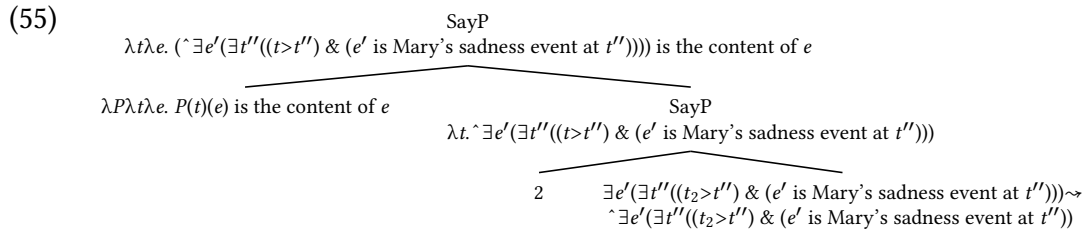
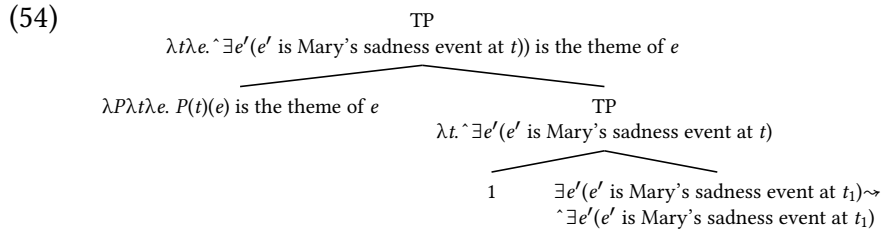
In (52), non-indexical present emerging via *PAST*-deletion is handled as t_k variable in the T head, which gets combined with an operator taking time variables and returning sets of events. Since the outcome is a set of events, it can combine with the set of events denoted by the verb; in (52), this is a set of seeing events. (The end result is abbreviated as indicated by the second line.) Finally, we make an existential closure over events.



In this paper, the *PAST* t_k complexes are handled as quantifiers over times, which need to undergo quantifier raising. In (53), we can make an abstraction over the index of the time variable left behind in the T-internal position, after which we have a function from times to a set of events. The *PAST* operator takes a time-variable (here t_2) and a function from times to a set of events, and returns a function from times to a set of events. Finally, we make an existential closure over times and events.

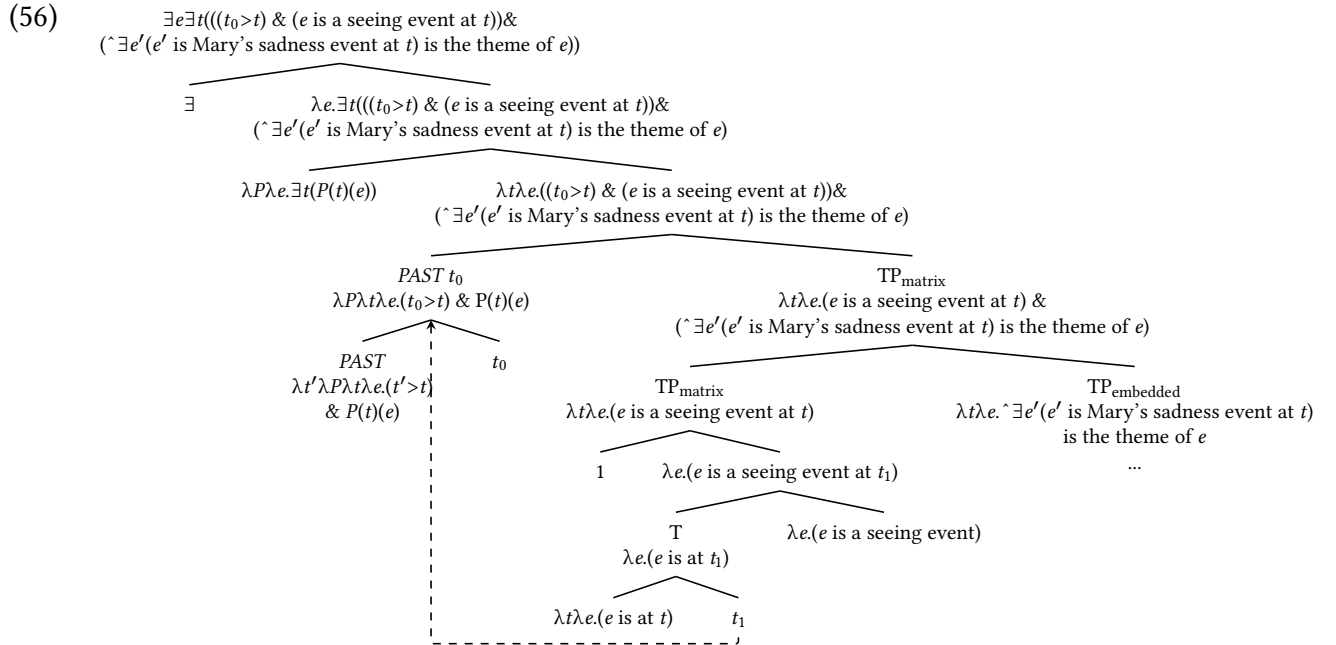


Compared to the previous subsection, the only change in the encoding of theta roles, theme in (54) and content in (55), take functions from times to propositions:



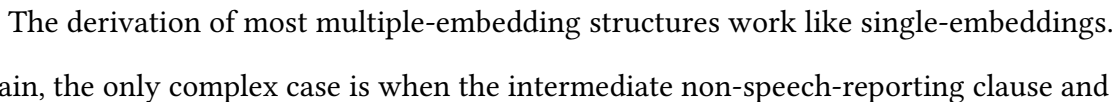
The tree in (56) shows the remaining part of the embedding of a non-speech reporting clause, that is, the LF-interpretation of the syntactic structure in (38). Again, this is the configuration in which only the matrix *PAST*-operator is handed to LF because the embedded *PAST* operator gets deleted by CSOT. In (56), the adjunction of (54) to the matrix TP happens right below the landing site of the quantifier raising of the matrix *PAST*-complex because we want the matrix time to serve as a reference time for the embedded

time. Since the quantifier raising is preceded by abstraction over times in the matrix clause, the matrix clause denotes a function from times to a set of events, which is the same type as the type of the topmost node in (54), that is, the two clauses can combine via Predicate Modification. Essentially, the t variables in the matrix clause is bound by the same lambda operator as the embedded time, that is, the matrix and the embedded clauses are *simultaneous*. The *PAST* operator takes t_0 as a reference time (which is “our now”), and then takes the matrix-embedded complex. In short, the matrix clause is backshifted relative to “our now”, while the embedded clause is not backshifted relative to the matrix attitude holder’s “now”. After this, we only need to make an existential closure over times and another over events. For reasons of simplicity, I ignore the fact that the matrix attitude can have a *mistaken perspective* (i.e. they are incorrect about “now”). Amending the system to capture the perspective facts is straightforward (Moulton 2009) but quite complex and not necessary for my point.



In structures involving speech-reporting clauses as (40), the embedded *PAST* does not get deleted, that is, it occurs in the LF in (57). In (57), the embedded clause derived in (55)

(57)



the deepest speech-reporting clause both attach to the matrix clause as in the syntactic derivation in (44) and in the event-semantic derivation in (51). The derivation of such double-embeddings is shown in (58). The general idea behind this derivation is that the existential operator over events undergoes raising from the intermediate clause to the edge of the matrix clause. This follows the idea seen in the previous section for (51): After the matrix set of events is closed over existentially, we can abstract over Index 4 and modify the set of events denoted by the intermediate clause. Informally, once we have worked out the relation between the intermediate and matrix clause, we can turn to the relation between the intermediate clause and the deepest clause. With the inclusion of times, the only complication is that the matrix time needs to serve as a reference time twice: Once for the matrix and the intermediate clause and once for the deepest clause. Consequently, the movement of the matrix *PAST*-complex happens in two steps: Once to above the intermediate clause and once above the deepest clause. The end result is that the matrix seeing time is backshifted relative to “our now”, the intermediate clause’s shouting is simultaneous with the matrix clause’s seeing, whereas the deepest clause is backshifted relative to the intermediate clause’s shouting and the matrix clause’s seeing.

(58)

